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(54) **RESPIRATOR APPARATUS**

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(57) **ABSTRACT**

One variation of a respirator apparatus includes a shield center section, a filter assembly, and a gasket. The shield center section defines: a transparent center region configured to align with a nose and a mouth of a user; and a filter window laterally offset from the transparent center region. The filter assembly includes a filter media and a seal: encircling a perimeter of the filter media; and defining a center groove configured to mate with a rim of the filter window to transiently retain the filter assembly within the filter window. The gasket is arranged on a rear face of the lens center section and configured to: seal against a face of the user; and locate the lens center section and the filter assembly offset from the face of the user to form an airflow channel to direct airflow across a glabrous cheek region of the user.

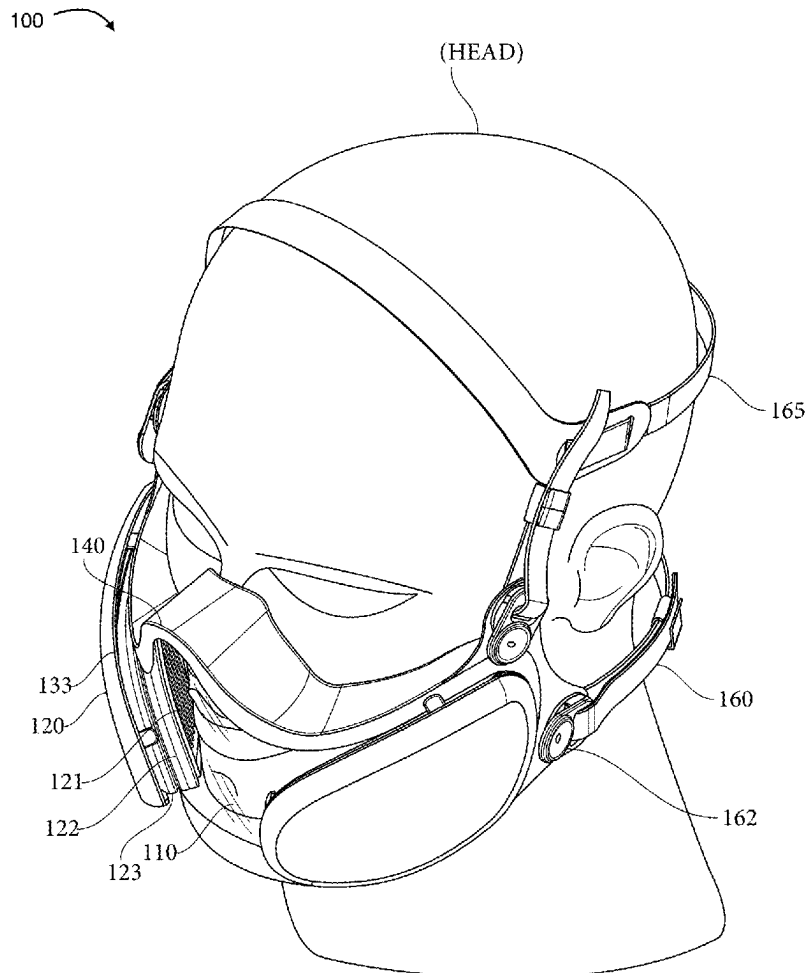
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(63) Continuation of application No. 18/520,316, filed on Nov. 27, 2023, now Pat. No. 12,337,200.

(60) Provisional application No. 63/428,314, filed on Nov. 28, 2022.



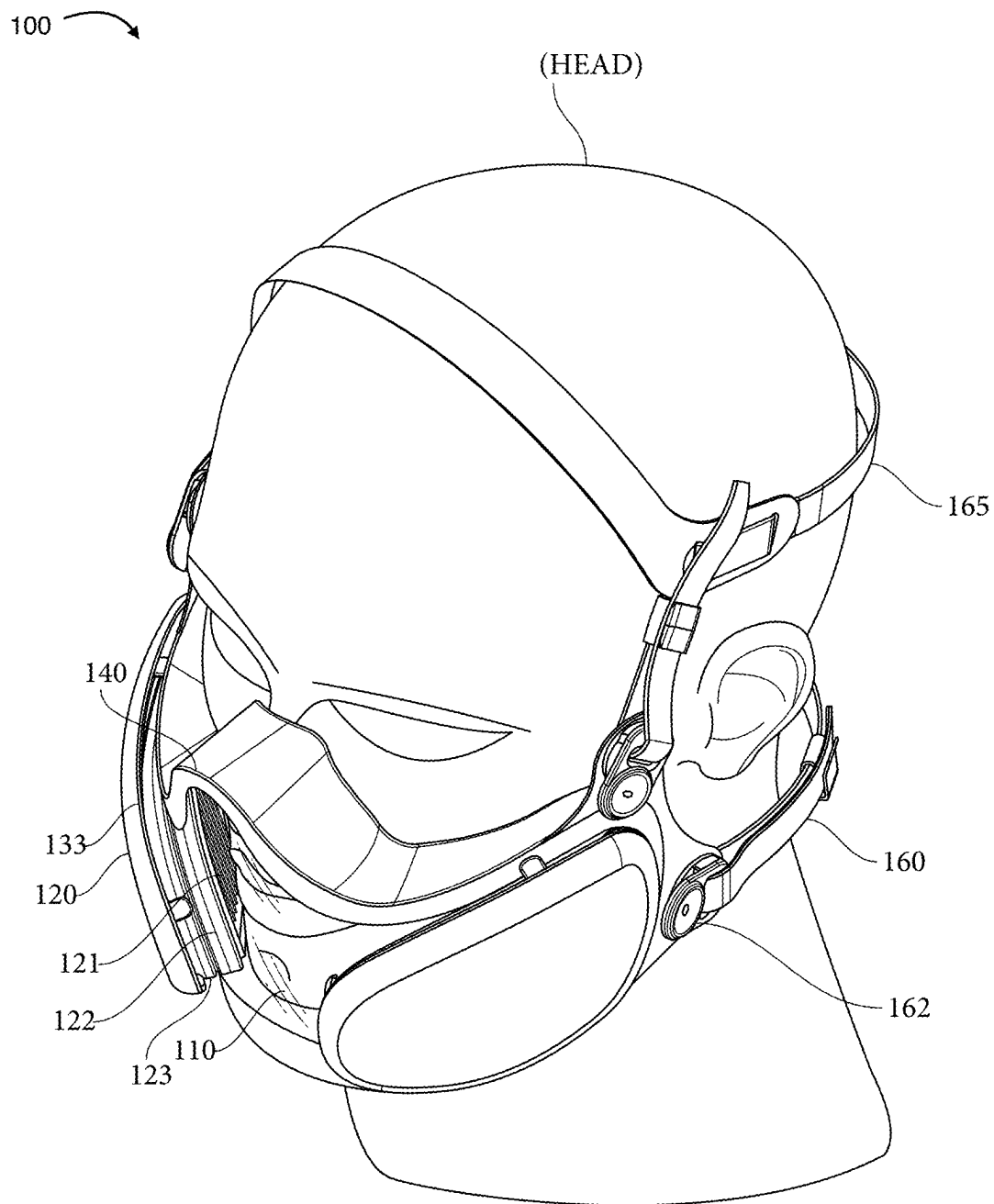


FIGURE 1

100

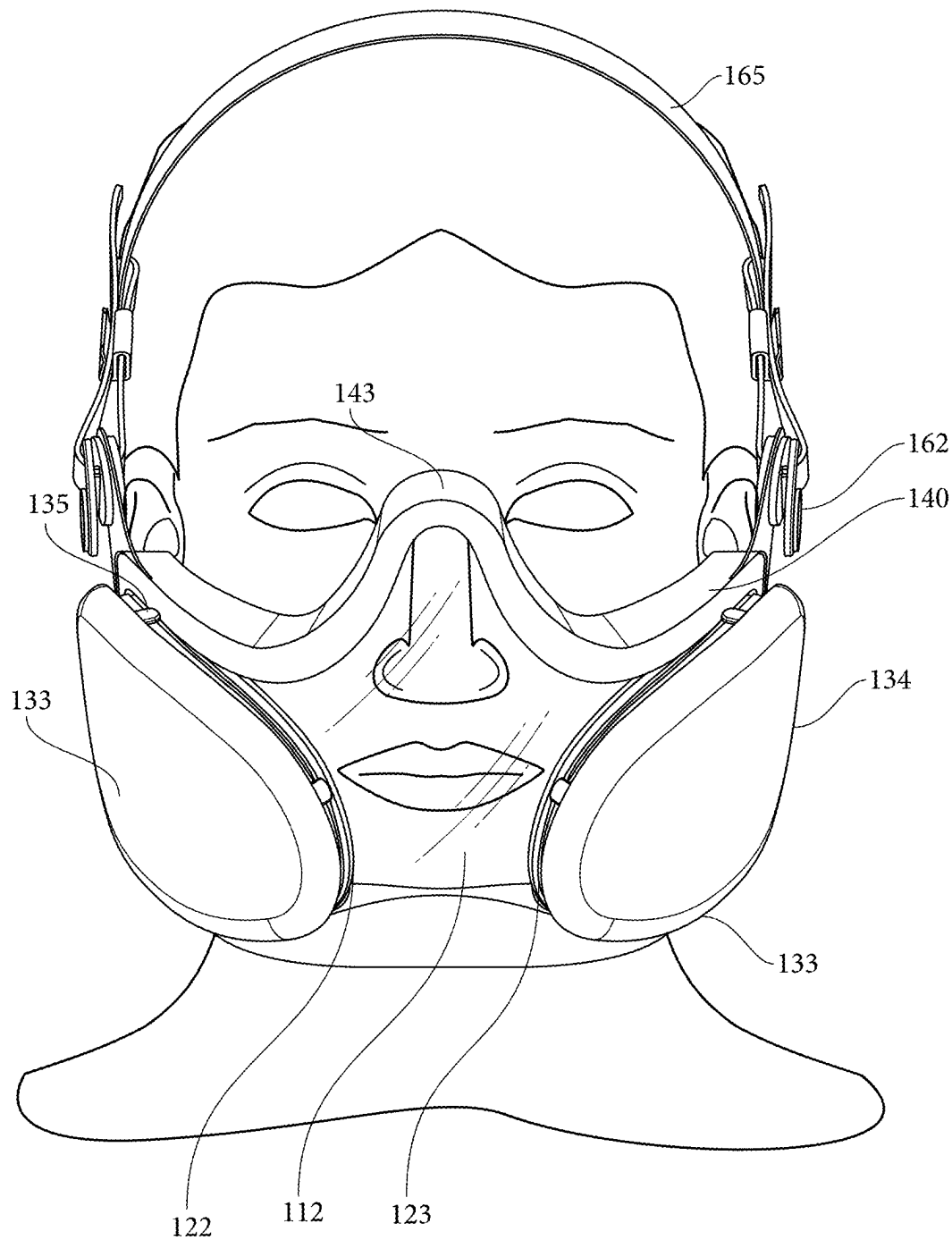


FIGURE 2

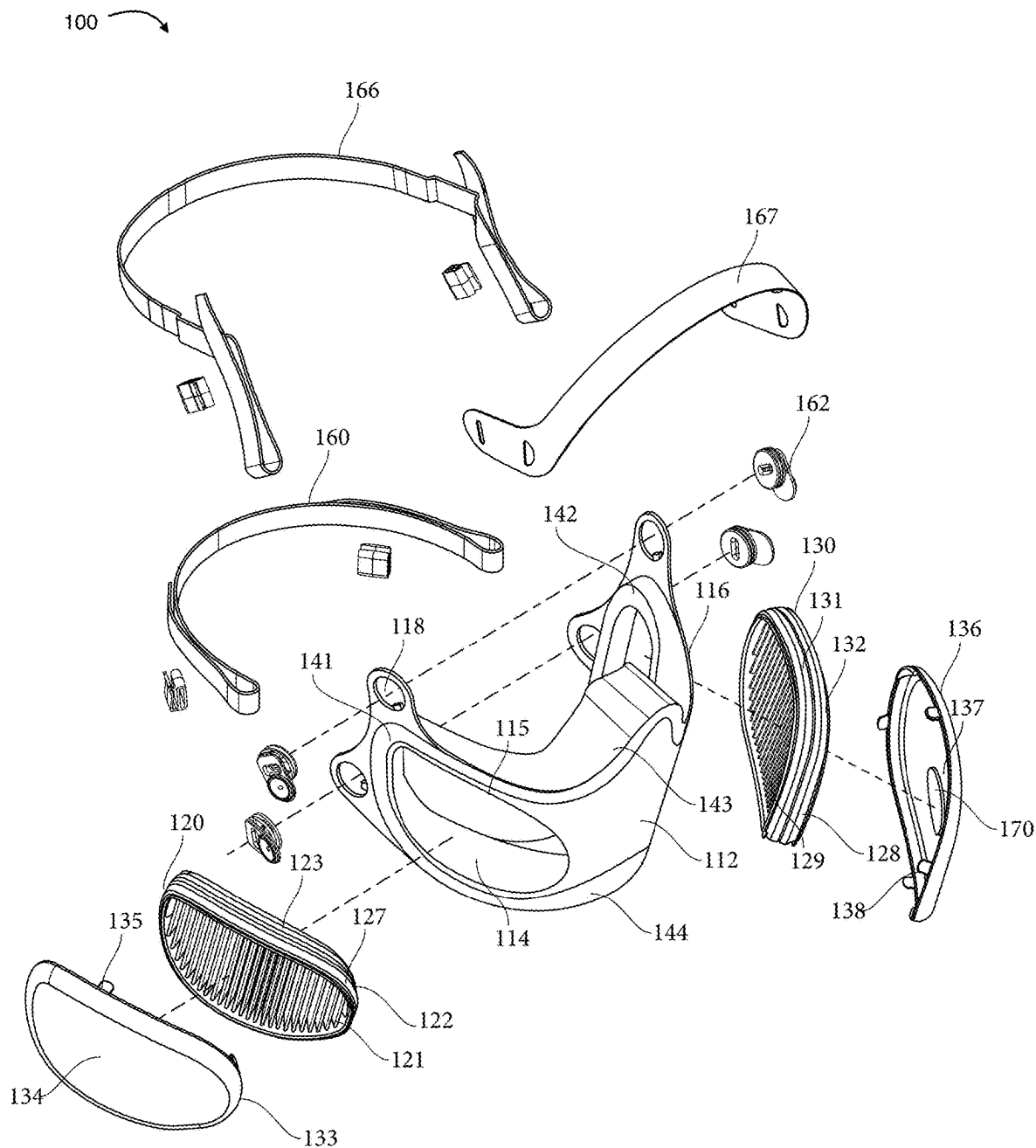


FIGURE 3

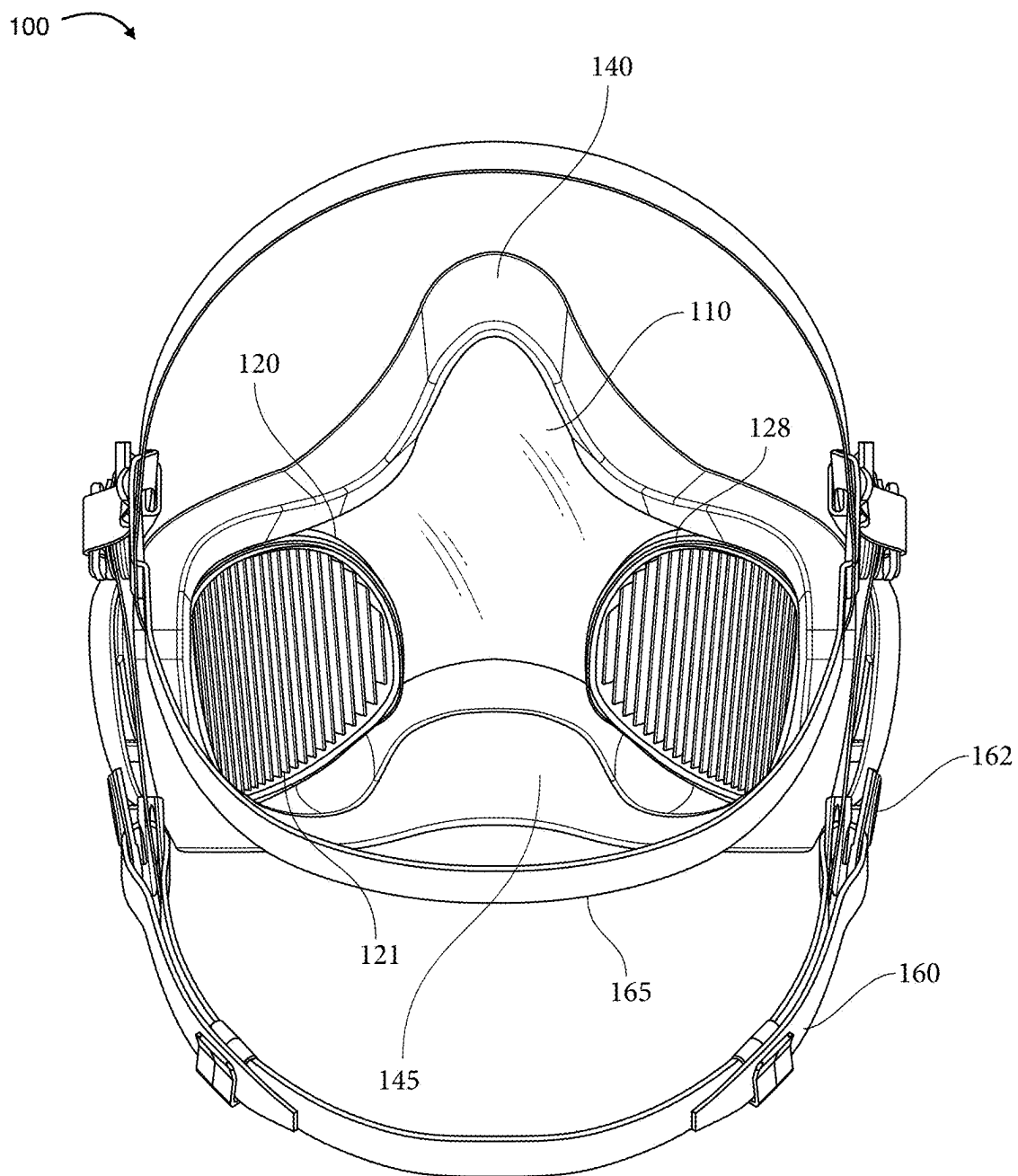


FIGURE 4

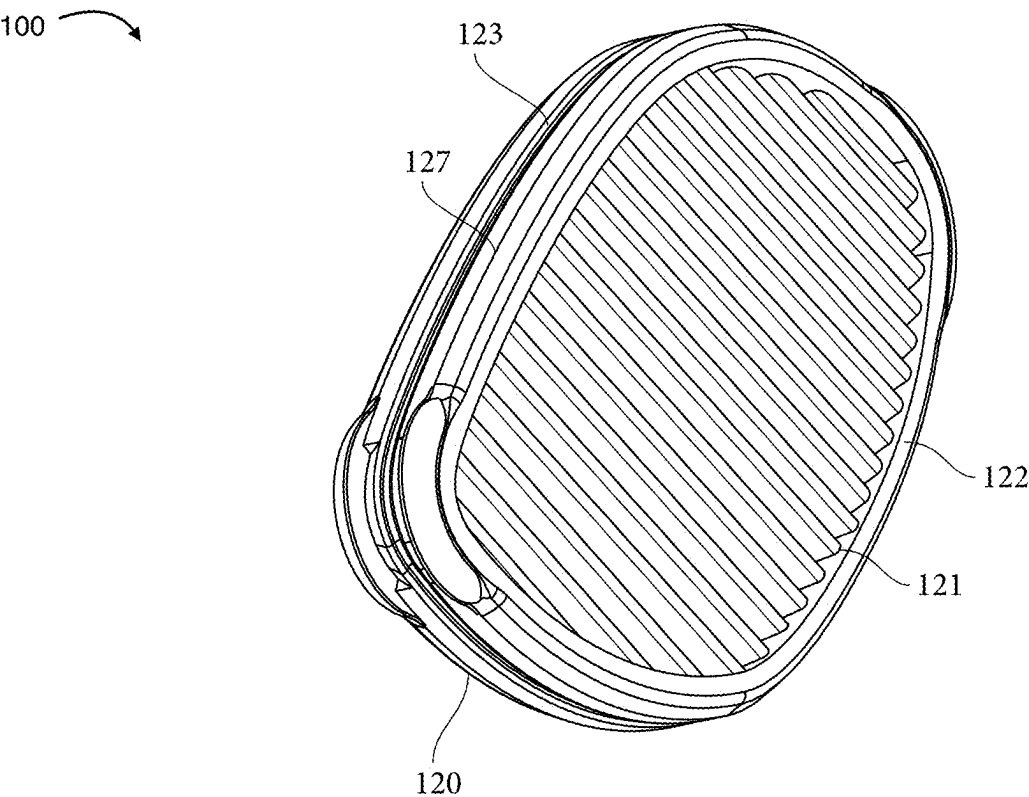


FIGURE 5

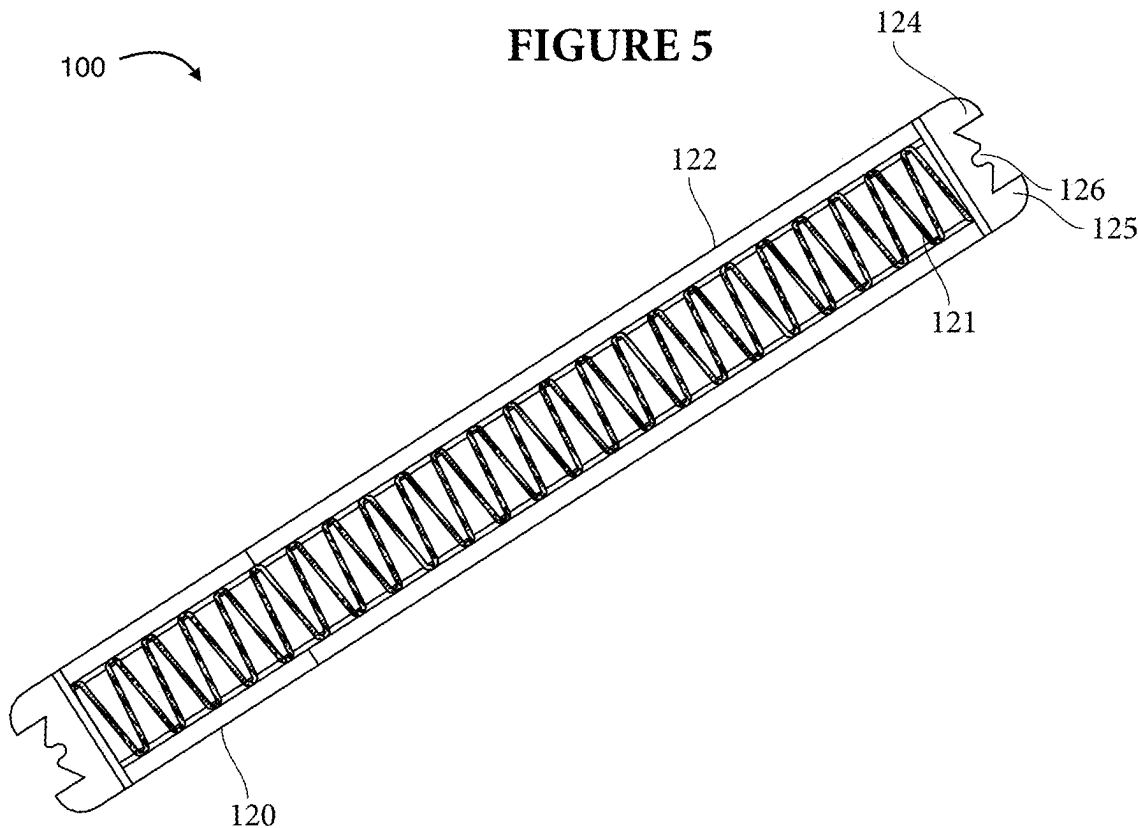


FIGURE 6

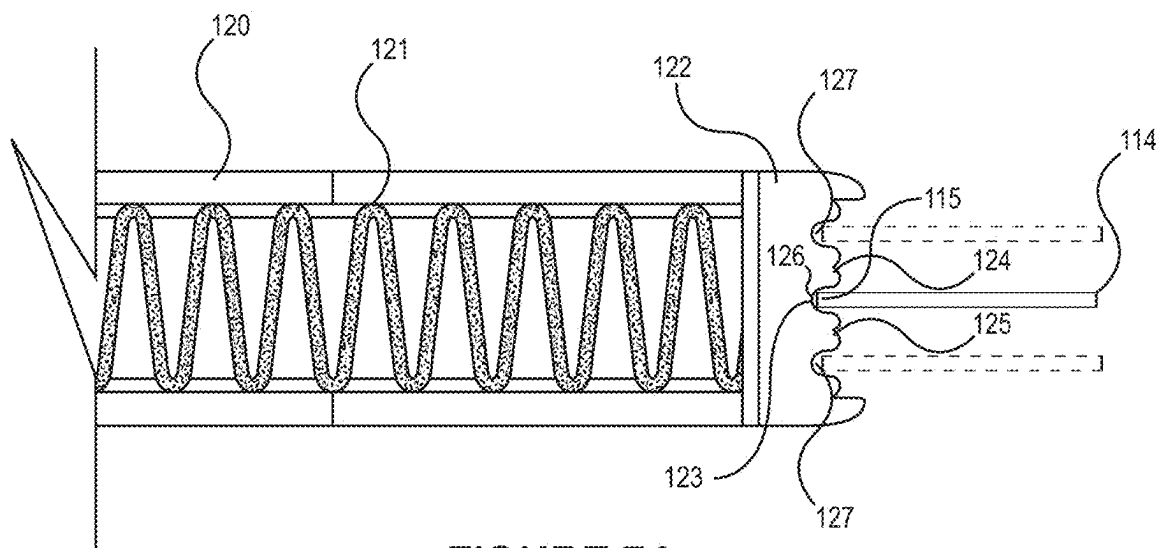


FIGURE 7A

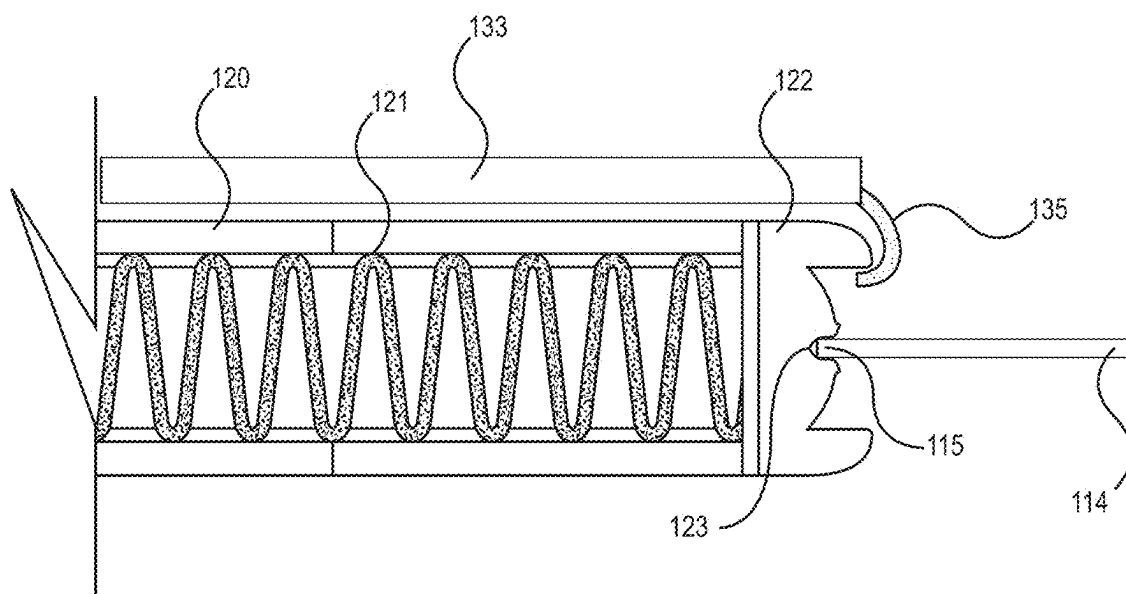
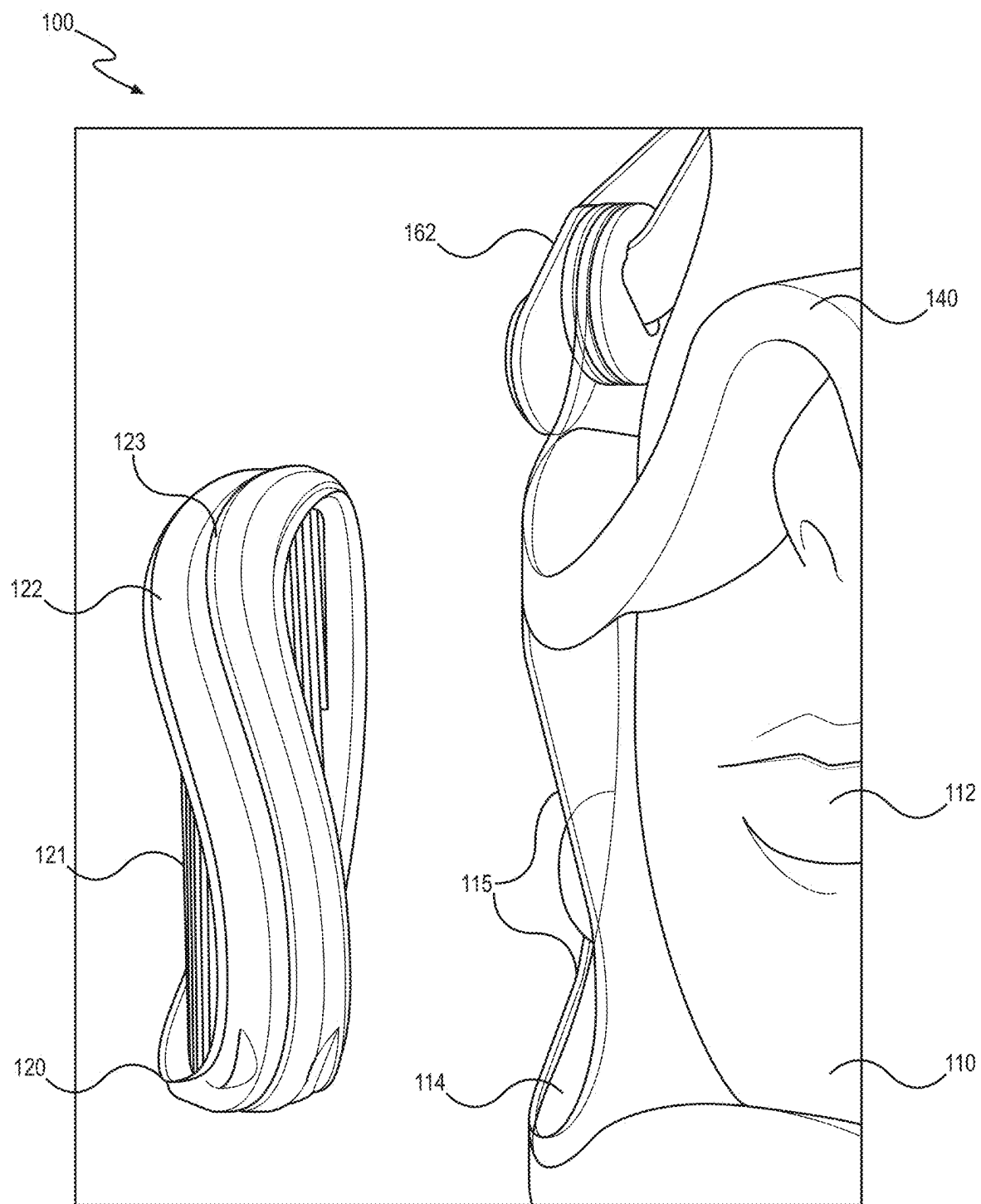
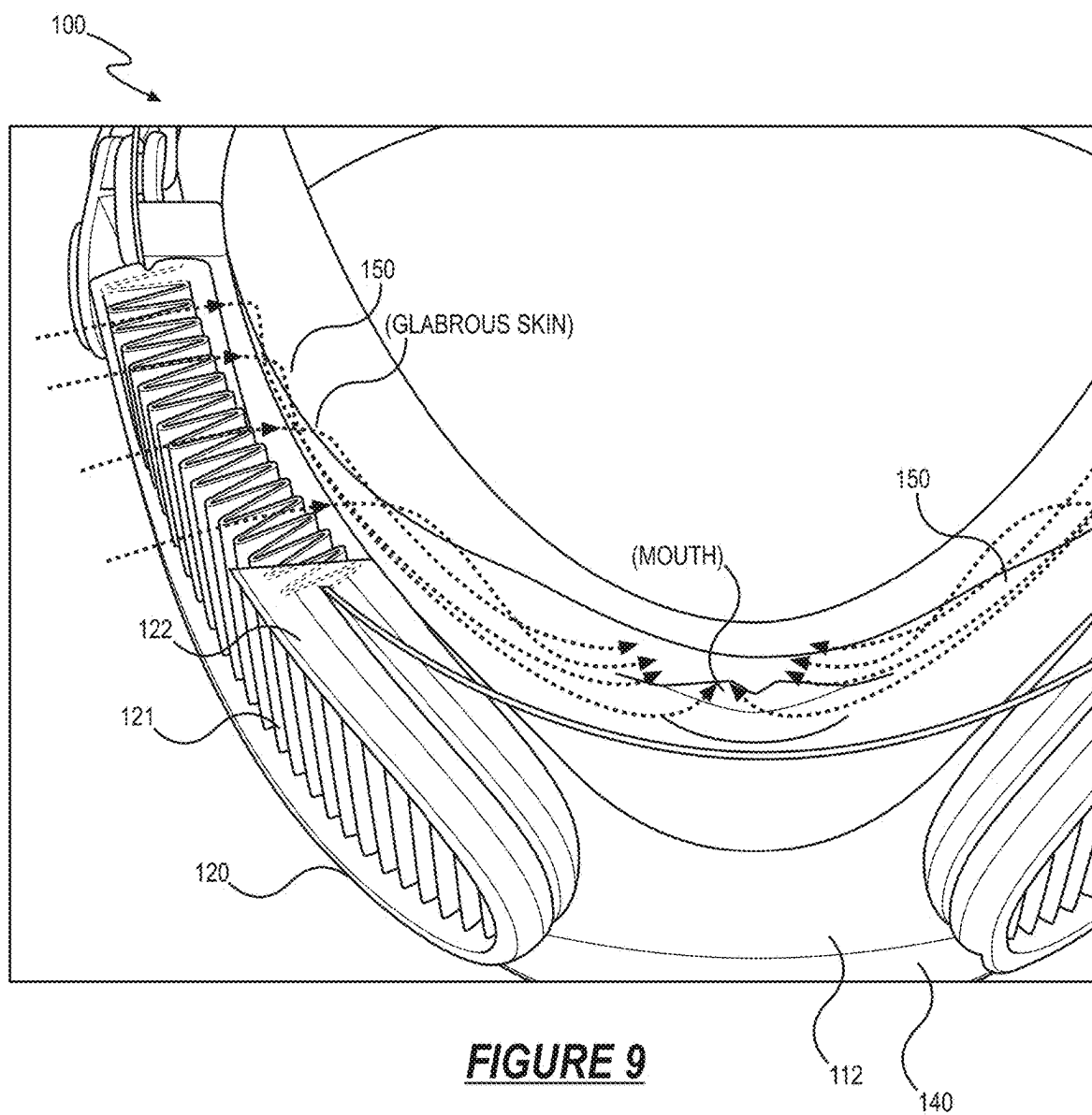


FIGURE 7B

**FIGURE 8**



RESPIRATOR APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Non-Provisional application Ser. No. 18/520,316, filed on 27 Nov. 2023, which claims the benefit of U.S. Provisional Application No. 63/428,314, filed on 28 Nov. 2022, each of which is hereby incorporated in its entirety by this reference.

TECHNICAL FIELD

[0002] This invention relates generally to the field of respirators and, more specifically, to a new and useful apparatus for a transparent shield respirator with conformal foam facial gasket incorporating interchangeable voice translucent, conformal frame pleated filters in the field of respirators.

BRIEF DESCRIPTION OF THE FIGURES

[0003] FIG. 1 is a schematic representation of a respirator apparatus;
 [0004] FIG. 2 is a schematic representation of the respirator apparatus;
 [0005] FIG. 3 is a schematic representation of the respirator apparatus;
 [0006] FIG. 4 is a schematic representation of the respirator apparatus;
 [0007] FIG. 5 is a schematic representation of the respirator apparatus;
 [0008] FIG. 6 is a schematic representation of the respirator apparatus;
 [0009] FIGS. 7A and 7B are schematic representations of the respirator apparatus;
 [0010] FIG. 8 is a schematic representation of the respirator apparatus; and
 [0011] FIG. 9 is a schematic representation of the respirator apparatus.

DESCRIPTION OF THE EMBODIMENTS

[0012] The following description of embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention. Variations, configurations, implementations, example implementations, and examples described herein are optional and are not exclusive to the variations, configurations, implementations, example implementations, and examples they describe. The invention described herein can include any and all permutations of these variations, configurations, implementations, example implementations, and examples.

1. Respirator Apparatus

[0013] As shown in FIG. 1, a respirator apparatus 100 includes: a shield center section 110; a first filter assembly 120; a second filter assembly 128; and a gasket 140.

[0014] The shield center section 110 is formed of a flexible material and is transparent in its entirety defining a center region 112 configured to locate centered over a face of a user; a left filter window 114 laterally offset from the transparent center region 112; and a right filter window 116 laterally offset from the transparent center region 112 opposite the left filter window 114.

[0015] The first filter assembly 120 includes a first filter media 121 and a first seal 122. The first seal 122: encircles a periphery of the first filter media 121; and defines a first center groove 123 configured to mate with a first rim 115 of the left filter window 114 to transiently retain the first filter assembly 120 within the left filter window 114.

[0016] The second filter assembly 128 includes a second filter media 129 and a second seal 130. The first seal 122: encircles a periphery of the second filter media; and defines a second center groove 131 configured to mate with a second rim of the right filter window 116 to transiently retain the second filter assembly 128 within the right filter window 116.

[0017] The gasket 140: is arranged on a rear face of the shield center section 110; and configured to conform with the contours of the face of the user and to center the transparent unobstructed window of the center region 112 with a nose and a mouth of the user. The gasket 140 is also configured to locate the shield center section 110 and the first filter assembly 120 offset from the face of the user to form a left airflow channel 150: defining a first height; defining a first depth less than the first height; and extending across a left cheek region of the user exposing glabrous skin regions on the upper cheek. The gasket 140 is also configured to locate the shield center section 110 and the second filter assembly 128 offset the face of the user to form a right airflow channel 150: defining a second height; defining a second depth less than the second height; and extending across a right cheek region of the user exposing glabrous skin regions on the upper cheek.

2. Applications

[0018] Generally, the respirator apparatus 100 can function as a respirator that: induces a cooling sensation across glabrous skin areas of the upper cheek region of the user; maintains contact with the face of the user by conforming responsively to movement of the face (e.g., head turning, speaking) by the user, demonstrating elasticity; and enables other users to view facial cues (e.g., expressions) from the user. More specifically, the respirator apparatus 100 can include: a shield center section 110 including a transparent region 112 configured to center over a nose and/or a mouth of the user; a filter assembly 120 transiently coupled to the shield center section 110 and arranged offset the transparent region 112 of the shield center section 110; and a gasket 140 arranged on a rear side of the shield center section 110 and configured to conformally mate with the face of the user to locate the filter assembly 120 proximal the glabrous cheek region of the user and form an airflow channel 150 within an interior volume of the shield center section 110 between the filter assembly 120 and the transparent center region 112 of the shield center section 110 identified as the plenum.

[0019] Thus, during inhalation (e.g., nasal inhalation, oral inhalation) and exhalation (e.g., nasal exhalation, oral exhalation) by the user, airflow within the airflow channel 150 is directed across the glabrous skin on the upper cheek region to induce the cooling sensation across the face of the user while maintaining an unobstructed view of the nose and the mouth of the user within the transparent center region 112 of the shield center section 110.

[0020] The respirator apparatus 100 includes the gasket 140: formed of an elastomeric conformal closed cell foam material including a non-porous skin surface; arranged about a periphery of the shield center section 110; including an

upper gasket section **143** defining an arched geometry contoured along an upper edge of the transparent center region **112** of the shield center section **110**; and including a lower gasket section **144** defining a concave geometry contoured along a lower edge of the transparent center region **112** of the shield center section **110**. Additionally, the gasket **140** is configured to: conform against the face of the user to form a plenum airflow channel **150** extending between the filter assembly **120** and the transparent center region **112** within the interior of the shield center section **110**; contour along a nasal bone of the user to maintain the upper gasket section **143** in contact with the face of the user; and contour along the mandible of the user to maintain the lower gasket section **144** in contact with the face of the user.

[0021] The respirator apparatus **100** can further include: a neck harness **160** coupled to the shield center section **110** and configured to wrap about a rear neck of the user; and a cranial harness **165** coupled to the shield center section **110** and configured to form a halo configuration about the cranium of the user. Accordingly, the neck harness **160** cooperates with the cranial harness **165** to: stabilize the gasket **140** about the face of the user to form the airflow channel **150** within the shield center section **110**; locate the nose and the mouth of the user within the transparent center region **112** of the shield center section **110**; and locate the filter window **114**—and therefore the filter assembly **120**—proximal the glabrous cheek region of the user.

[0022] Thus, the gasket **140** cooperates with the face of the user such that—rather than the gasket **140** sliding across the face of the user during movement of the face (e.g., head turns, mandible movement during speaking) by the user—the gasket **140** deflects, stretches, and compresses to conform to facial topology demonstrating homogenous elasticity, thereby: maintaining contact between the upper gasket section **143** and the nasal bone of the user; maintaining contact between the lower gasket section **144** and the bottom side of the mandible of the user; and directing airflow through the filter assembly within the interior volume of the shield center section **110**.

[0023] Additionally, the shield center section **110** further includes a filter window **114**: offset from the transparent center region **112** and configured to receive the filter assembly **120**; and configured to receive the filter assembly **120** to locate the airflow channel **150** proximal the glabrous cheek region of the user. In particular, the filter assembly **120** can include: a filter media **121** (e.g., polytetrafluoroethylene membrane, spunbonded polyester); and a seal **122** encircling the filter media **121** and defining a center groove **123** extending about the periphery of the seal **122** and configured to mate with a rim **115** of the filter window **114** to locate the filter assembly **120** within the filter window **114**. The seal **122** can: be formed of an elastomeric material configured to deform during installation of the filter assembly **120** into a filter window **114**; and include a groove configured to expand around and to seal against inner and outer edges of a rim **115** of the filter window **114**, thereby locating and sealing the filter assembly **120** within the filter window **114**. Accordingly, the filter assembly **120**: is operable in a deformed configuration by the user to locate the filter assembly **120** within the filter window **114** of the shield center section **110**; and is operable in a nominal configuration to locate the rim **115** of the filter window **114** within the center groove **123** of the seal **122** and thus form an interference fit between the filter window **114** and the filter

assembly **120** that directs airflow from an exterior side of the shield center section **110** through the filter assembly **120** within the interior volume of the shield center section **110**. Thus, during inhalation by the user (e.g., from the nose, mouth), airflow within the airflow channel **150** is directed from the filter assembly **120** across the glabrous cheek region of the user to induce the cooling sensation. Additionally, during exhalation by the user (e.g., from the nose, mouth), airflow within the airflow channel **150** is directed from the transparent center region **112** across the glabrous cheek region of the user to induce the cooling sensation.

[0024] Therefore, the respirator apparatus **100**: induces a cooling effect across the glabrous cheek region of the user during inhalation and exhalation of the user; enables an outside observer to view facial cues from the nose and the mouth region of the user located within the transparent center region **112** of the shield center section **110**; deflects and twists about the face of the user to maintain the gasket **140** against the face of the user to direct airflow from the filter assembly **120** toward the interior volume of the shield center section **110**; and prevents contaminants (e.g., airborne particulates, dirt, dust) about an exterior environment of the shield center section **110** from entering the interior of the shield center section **110** identified as the plenum.

3. Lens Section

[0025] Generally, the respirator apparatus **100** includes a shield center section **110**: formed of an transparent flexible material (e.g., polycarbonate film) configured to deflect and twist responsive to movement of the face by the user; and configured to contour a face of the user to expose a nose and a mouth region of the user. More specifically, the respirator apparatus **100** includes: a transparent center region **112** configured to locate over a face (e.g., nose, mouth) of the user; a left filter window **114** laterally offset from the transparent center region **112**; and a right filter window **116** laterally offset from the transparent center region **112** opposite the left filter window **114**. Thus, the respirator apparatus **100**: exposes the nose region and the mouth region to an observer when arranged on the face of the user; and supports transient coupling of a filter assembly **120** at the left filter window **114** and/or the right filter window **116** to prevent contaminants (e.g., airborne particulates, dirt, dust) about an exterior environment of the lens center section **110** from entering the interior of the shield center section **110**.

[0026] In one implementation, the shield center section **110**: defines a target width configured to curve from the left zygomatic bone to the right zygomatic bone of the user to locate the shield center section **110** to across a breathing zone (e.g., mouth, nose) and glabrous cheek regions of the user; and defines a target height configured to extend from a bridge of the nose to below a chin of the user. The shield center section **110** includes: an upper edge defining an upper arch configured to contour along a nose bone of the user; a lower edge defining a lower arch configured to contour along a mandible of the user; a left edge interposed between the upper edge and the lower edge and configured to contour along a left zygomatic bone of the user; and a right edge, opposite the left edge, interposed between the upper edge and the lower edge and configured to contour along a right zygomatic bone of the user. More specifically, the upper edge can further define: a left scalloped region configured to contour along a left maxilla bone, below a left eye, of the user to form an unobstructed field of view from the left eye

when locating the lens center section 110 on the face of the user; and a right scalloped region—opposite the left scalloped region—configured to contour along a right maxilla bone, below the right eye, of the user to form an unobstructed field of view from the right eye when locating the shield center section 110 on the face of the user.

[0027] In this implementation, the shield center section 110 includes the left filter window 114: arranged proximal the left side of the lens center section 110; defining a target shape (e.g., semi-elliptical shape, circular shape, lozenge shape); defining a window width; and defining a window height less than the window width. Similarly, the shield center section 110 includes the right filter window 116: arranged proximal the right side of the shield center section 110; defining the target shape (e.g., semi-elliptical shape, circular shape, lozenge shape); defining the window width; and defining the window height less than the window width. Furthermore, the filter window 114 can include indicia (e.g., contrasting color band) arranged about the periphery of the filter window 114 representing a sealing edge configured to guide a user to locate the filter assembly 120 within the filter window 114 of the shield center section 110.

[0028] Although this implementation includes a left filter window 114 and a right filter window 116 arranged on the shield center section 110, other variations can include a single filter window and/or multiple (e.g., greater than two) filter windows arranged on the shield center section 110 configured to receive the filter assembly 120.

[0029] Therefore, the shield center section 110: enables observers to view facial cues (e.g., facial expressions) from the user through the transparent center region 112; includes the filter assembly 120 transiently coupled within the right filter window 116 and the left filter window 114 of the shield center section 110; and flexes and twists, responsive to movement of the face by the user.

4. Filter Assembly

[0030] Generally, the filter assembly 120 includes: a filter media 121; and a seal 122. The filter media 121: is configured to prevent contaminants (e.g., airborne particulates, dirt, dust) from passing from an exterior environment outside of the lens center section 110 into the interior volume of the respirator apparatus 100 between the shield center section 110 and the user's face. The seal 122: is arranged about (e.g., overmolded) the filter media 121; and defines an outer circumferential groove configured to seat around and compress against the rim 115 of a filter window 114 to thus form a double-seal against the inner and outer edges of the rim 115 of the filter window 114 while locating the filter media 121 within the filter window 114. More specifically, the filter assembly 120 is configured to deform-responsive to an applied force about the seal 122 encircling the filter media 121—to locate the filter assembly 120 entirely within the filter window 114. Accordingly, upon release of the applied force about the seal 122, the filter assembly 120 returns to a nominal shape (e.g., semi-elliptical shape) to induce mating between the center groove 123 arranged about the seal 122 and the rim 115 about the filter window 114, thereby forming the interference fit between the filter window 114 and the filter assembly 120. Thus, the filter assembly 120 can: block contaminants (e.g., airborne particulates, dirt, dust) from entering an interior of the shield center section 110; and form an interference fit between the filter assembly 120 and the filter window 114 to direct airflow from outside

the shield center section 110 through the filter assembly 120 into the interior volume of the shield center section 110.

4.1 Interference Fit

[0031] In one implementation, the respirator apparatus 100 includes: the filter window 114 (e.g., left filter window 114 and/or left filter window 114) defining a square rim 115 about a periphery of the filter window 114; and the center groove 123 defining a forked geometry extending about a periphery of the seal 122 encircling the filter media 121 and configured to mate with the square rim 115 of the filter window 114.

[0032] In one example, the forked geometry includes: a first tine 124 encircling the periphery of the seal 122; a second tine 125—opposite the first tine 124—encircling the periphery of the seal 122; and a slot 126 interposed between the first tine 124 and the second tine 125 configured to receive the square rim 115 of the filter window 114. Accordingly, the forked geometry of the center groove 123 extends about the periphery of the seal 122 to mate with interior and exterior edges of the rim (e.g., four-sided rim, triangular rim, circular rim) about the periphery of the filter window 114 to form the interference fit between the filter assembly 120 and the shield center section 110. In this example, the center groove 123 can transiently couple the filter window 114 to: locate the square rim 115 of the filter window 114 within the slot 126 of the forked geometry of the center groove 123; seal the first tine 124 of the forked geometry against exterior edges of the rim 115 of the filter window 114; and seal the second tine 125 of the forked geometry against interior edges, opposite the exterior edges, of the rim 115 of the filter window 114, thereby forming the interference fit between the filter assembly 120 and the filter window 114.

[0033] In one implementation, the filter assembly 120 is operable: during application of a force about the periphery of the filter assembly 120, in a deformed configuration to locate the filter assembly 120 within the filter window 114; and, during absence of an applied force about the periphery of the filter assembly 120, in a nominal configuration to seal the filter assembly 120 against the rim of the filter window 114.

[0034] In one example, the filter media 121: includes a multi-layer construction (e.g., single-layer or two-layer polytetrafluoroethylene membrane, where the polytetrafluoroethylene membrane layer is bonded on each side to spunbonded polyester scrim) of a target filtering efficiency (e.g., between 95% for R95, N95 filtering efficiency and 99.97% for P100, N100 filtering efficiency); defines a parallel pleated geometry enclosed within the seal 122 about the filter media 121; and spans a target surface area (e.g., between 80 square inches and 110 square inches, between 51000 square millimeters and 70967.6 square millimeters) within the seal 122. Additionally, the seal 122: is formed of an elastomeric material (e.g., urethane, thermoplastic) overmolded about the filter media 121; and defines a target durometer range (e.g., between 55 Shore A and 75 Shore A) configured to deform the filter assembly 120 responsive to an applied force about the seal 122 encircling the filter media 121.

[0035] In another example, the filter window 114 defines: a window width (e.g., between 3 inches and 4 inches); and a window height (e.g., between 0.5 and 1.0 inches) less than the window width. In this example, the filter assembly 120: defines a semi-elliptical geometry matching the window

width and the window height; and is configured to laterally and vertically deform responsive to an applied force about the periphery of the filter assembly 120. Accordingly, the user can then: apply a force about the seal 122 encircling the filter media 121 to operate the filter assembly 120 in a deformed configuration to locate the filter assembly 120 within the window width and the window height of the filter window 114; and remove the applied force about the seal 122 to operate the filter assembly 120 in the nominal configuration to mate the center groove 123 with the rim of the filter window 114, thereby forming the interference fit between the filter assembly 120 and the filter window 114. [0036] In one implementation, the filter assembly 120 can be interchanged across multiple filter windows located across the shield center section 110. For example, the shield center section 110 can include: a left filter window 114 arranged proximal the left side of the shield center section 110; and a right filter window 116 arranged proximal the right side of the shield center section 110. Accordingly, the filter assembly 120 is selectively arranged within the left filter window 114 or the right filter window 116 by the user. More specifically, the filter assembly 120: is operable in a first configuration including the center groove 123 of the seal 122 mating a first rim 115 of the left filter window 114 to transiently retain the filter assembly 120 within the left filter window 114; and is operable in a second configuration including the center groove 123 of the seal 122 mating a second rim of the right filter window 116, opposite the left filter window 114, to transiently retain the filter assembly 120 within the right filter window 116.

[0037] In another implementation, the respirator apparatus 100 includes a first filter assembly 120 including: a first filter media 121; and a first seal 122 encircling the first filter media 121 and defining a first center groove 123 configured to transiently couple the left filter window 114 of the shield center section 110. Similarly, the respirator apparatus 100 can include a second filter assembly 128 including: a second filter media 129; and a second seal 130 encircling the second filter media 129 and defining a second center groove 131 configured to transiently couple the right filter window 116 of the shield center section 110. In this implementation, as described above, the first filter assembly 120 and the second filter assembly 128 are interchangeable across the left filter window 114 and the right filter window 116. Additionally, although this implementation includes a first filter assembly 120 and a second filter assembly 128 configured to transiently couple the shield center section 110, other variations can include a single filter assembly and/or multiple (e.g., greater than two) filter assemblies configured to transiently couple filter windows arranged across the shield center section 110.

[0038] Therefore, the filter assembly 120 transiently couples the filter window 114 of the shield center section 110 in order to: during inhalation by the user, prevent contaminants (e.g., airborne particulates, dirt, dust) from entering an interior volume of the shield center section 110; and form an interference fit between the filter assembly 120 and the filter window 114 to direct airflow from outside the shield center section 110 through the filter assembly 120 toward the interior volume of the shield center section 110.

4.2 Filter Covers

[0039] In one implementation, the filter assembly 120 includes a seal 122: encircling the periphery of the filter

media 121; defining a center groove 123 extending about the periphery of the seal 122; and defining an intermittent outer groove 127—adjacent the center groove 123—encircling the periphery of the seal 122 and configured to transiently couple a filter cover 133 extending across the filter media 121 of the filter assembly 120. In this implementation, the filter cover 133 includes: a pre-filter media 134 (e.g., sili-conized, Sil-Scrim) heat molded polyester scrim, carbon scrim); and a capture detent 135 arranged about an internal periphery of the pre-filter media 134 and configured to couple the outer intermittent groove 127 to transiently retain the filter cover 133 across an outer face of the filter media 121. Accordingly, during retention of the filter cover 133 across the outer face, the filter cover 133 cooperates with the filter assembly 120 to form a plenum between the surface area of the filter media 121 of the filter assembly 120 and the pre-filter media 134 of the filter cover 133.

[0040] Additionally, the filter cover 133 can include a valve plate 170 configured to mate to the filter rim geometry: centrally located on the pre-filter media 134; arranged within the plenum formed by the filter assembly 120 and the filter cover 133; and configured to avoid inward airflow during inhalation by the user and outward airflow during exhalation by the user. Thus, the filter cover 133 cooperates with the filter assembly 120 to isolate the filter media 121 of the filter assembly 120 from an external environment in order to phase airflow (e.g., inward airflow, outward airflow) and prevent large particulates and liquid contaminants (e.g., >PM10 airborne particulates, dirt, dust, body fluids, blood, water droplets) from directly contacting the filter media 121 of the filter assembly 120.

[0041] Additionally, the valve plate 170, centrally located on the pre-filter media 134 suspended proximal to the inlet opening of the filter assembly 120 separated from the filter assembly 120 inlet face by the plenum as to prevent interference with inward airflow during inhalation or outward airflow during exhalation. The valve plate 170 is configured to cover the inlet face of the filter assembly 120. The configuration of the pre-filter media 134 incorporates flexibility of the outer plane of the pre-filter media 134 that suspends the valve plate 170 proximal to the face of the filter assembly 120. Upon demand the valve plate 170 can be deflected by force to engage the face of the filter assembly 120 and obstruct inlet and outlet airflow to allow an evaluation of the gasket 140 seal mating integrity to the face of the user. The force to engage the valve plate 170 to the face of the filter assembly 120 is less than 1 inch-pound so as not to induce a false positive gasket seal mating integrity evaluation by adding pressure to the gasket 140 seal during the applied pressure sequence of the evaluation process. The evaluation requires short and abrupt inhalation bursts sufficient to generate a vacuum in the plenum of the respirator while the valve plates 170 are simultaneously deflected and pressed against the faces of the filter assemblies 120. The resulting increase in vacuum caused by an abrupt inhalation draws the respirator apparatus 100 toward the face increasing pressure from the gasket 140 onto the facial sealing land readily perceived by the user. Weak or lack of vacuum during the evaluation indicates gasket seal leakage at the gasket-face sealing land and requires refitting the respirator apparatus 100 until a perceptible vacuum can be performed.

[0042] In one example, the respirator apparatus 100 includes a first filter assembly 120 including a first seal 122 defining: a first center groove 123 extending about the

periphery of the first seal **122**; and a first intermittent outer groove **127**—adjacent the first center groove **123**—encircling the periphery of the first seal **122**. In this example, the first filter assembly **120** can transiently couple the left filter window **114** to locate the first outer groove **127** at an exterior side of the lens center section **110**. Additionally, the respirator apparatus **100** includes a first filter cover **133** including: a first pre-filter media **134**; and a first capture detent **135** arranged about a periphery of the first pre-filter media **134** and configured to couple the first intermittent outer groove **127** to transiently retain the first filter cover **133** across a first outer face of the first filter media **121** of the first filter assembly **120**. Similarly, the respirator apparatus **100** includes a second filter assembly **128** including a second seal **130** defining: a second center groove **131** extending about the periphery of the first seal **122**; and a second intermittent outer groove **132**—adjacent the second center groove **131**—encircling the periphery of the second seal **130**. The second filter assembly **128** can transiently couple the right filter window **116** to locate the second outer groove **132** at the exterior side of the lens center section **110**. Furthermore, the respirator apparatus **100** includes a second filter cover **136** including: a second pre-filter media **137**; and a second capture detent **138** arranged about a periphery of the second pre-filter media **137** and configured to couple the second intermittent outer groove **132** to transiently retain the second filter cover **136** across a second outer face of the second filter media **129** of the second filter assembly **128**.

[0043] Therefore, the respirator apparatus **100** can include a filter cover **133** configured to transiently couple the filter assembly **120** in order to: prevent large particulate and liquid contaminants (e.g., >PM10 airborne particulates, dirt, dust, body fluids, blood, water droplets) from directly contacting the filter media **121** of the filter assembly **120**; and direct airflow (e.g., inward airflow, outward airflow) during inhalation and exhalation by the user.

4.3 Adjustable Filter Depth

[0044] In one implementation, the filter assembly **120**: is operable in a first depth configuration including the center groove **123** of the seal **122** coupling the filter window **114** to locate the filter assembly **120** at a first depth (e.g., between 0.1 and 0.2 inches) within the interior volume of the lens center section **110**; and is operable in a second depth configuration including the outer groove **127** of the seal **122** coupling the filter window **114** to locate the filter assembly **120** at a second depth, greater than the first depth (e.g., between 0.21 and 0.3 inches), within the interior volume of the lens center section **110**. More specifically, the filter assembly **120**: is operable in the first depth configuration including the center groove **123** of the seal **122** mating with the rim of the filter window **114** to transiently retain the filter assembly **120** within the filter window **114** at the first depth; and is operable in the second depth configuration including the outer groove **127** of the seal **122** mating with the rim of the filter window **114** to transiently retain the filter assembly **120** within the filter window **114** at the second depth greater than the first depth.

[0045] Therefore, the filter assembly **120** is selectively arranged within the filter window **114** to adjust depth of the filter assembly **120** extending within the interior volume of the lens center section **110** in order to prevent direct contact of the filter assembly **120** with the face of the user when larger filters are installed. Although, this implementation

includes the seal **122** defining a center groove **123** and an outer groove **127** about the periphery of the seal **122**, other variations can include a single groove and/or multiple (e.g., greater than two) grooves arranged about the periphery of the seal **122** encircling the filter media **121** of the filter assembly **120**.

5. Gasket

[0046] Generally, the respirator apparatus **100** includes a gasket **140**: arranged about a rear side of the lens center section **110**; and configured to conform against the face of the user. More specifically, the gasket **140**: is arranged (e.g., adhesively coupled, overmolded) about a periphery of the lens center section **110**; is configured to contour about the face of the user to locate the nose and mouth of the user within the transparent center region **112** of the shield center section **110**; and forms an airflow channel **150** within the interior volume of the shield center section **110** to direct airflow—during inhalation and exhalation by the user—across a glabrous cheek region of the user. Thus, the gasket **140**: during movement of the face by the user, maintains contact against the face of the user to prevent air from entering and/or exiting the interior volume (e.g., plenum) of the lens center section **110** along the gasket **140** compressed against the face of the user; and induces a passive cooling effect—during inhalation and exhalation by the user—across the face of the user as air passing through the airflow channel **150** flows across the glabrous cheek region of the user.

5.1 Facial Compression

[0047] In one implementation, the gasket **140**: is formed of an elastomeric closed cell conformal foam material including a non-porous surface (e.g., integrated skin); is arranged about the rear side of the shield center section **110**; and is configured to conform against the face of the user to locate the filter window **114** proximal the glabrous cheek region of the user. In particular, the gasket **140**: is formed of a elastomeric urethane foam and/or a elastomeric silicone foam material; includes a continuous smooth skin surface integrally formed during the specialized foam molding process that implements a proprietary tool coating that breaks surface tension between the tool contact surfaces and the foam, on contact, collapsing the air cells near the surface of the foam, forming a uniformly smooth skin surface that has increased durability, improved release characteristics and an ideal sealing land surface finish against facial contours; and defines a continuous boundary about the rear side of the lens center section **110** to contour the nose, cheeks, and chin of the user to maintain contact against the face of the user during movement of the face (e.g., movement of the mandible) by the user.

[0048] In this implementation, the gasket **140** includes a left gasket section **141**: defining a first depth (e.g., 0.25 inches, 6 millimeters) contoured along a left edge of the lens center section **110**; and configured to conform a left side of a mandible of the user to locate the left filter window **114** of the lens center section **110** proximal a left glabrous cheek region of the user. Similarly, the gasket **140** includes a right gasket section **142**: matching the first depth (e.g., 0.20 inches, 5 millimeters) contoured along a right edge of the lens center section **110**; and configured to conform against the right side of the mandible of the user to locate the right

filter window **116** of the lens center section **110** proximal a right glabrous cheek region of the user. The entire gasket **140** is a single continuous injection molded part designed to eliminate all mold parting lines that create potential leak paths between the sealing land of the gasket and facial topology.

[0049] Accordingly, the right gasket section **142** cooperates with the left gasket section **141** to: maintain contact between the gasket **140** and the face of the user (e.g., Sealing Land); and direct airflow through the filter assembly **120** into the interior volume of the lens center section **110** (e.g., Plenum). Thus, during movement of the face by the user (e.g., movement of the mandible, turning of the head), the gasket **140**—by the design of the gasket's elastomeric, foam conforming characteristics and geometry sculpted specifically to move with the gasket sealing land's interface on facial topology—maintains contact with the face of the user.

[0050] Additionally, the gasket **140** includes an upper gasket section **143**: defining a second depth greater than the first depth (e.g., 0.25 inches, 6 millimeters); defining an arched geometry contoured along an upper edge of the transparent center region **112**; and configured to conform against the skin supported by a nasal bone and a zygomatic bone of the user to locate the nose of the user within the transparent center region **112** of the shield center section **110**. Furthermore, the gasket **140** includes a lower gasket section **144**: defining third depth greater than the second depth (e.g., 0.80 inches, 20 millimeters); defining a concave geometry contoured along a lower edge of the transparent center region **112**; and configured to conform against a bottom side of the mandible of the user to locate a chin and the mouth of the user within the transparent center region **112** of the lens center section **110**.

[0051] Accordingly, the lower gasket section **144** cooperates with the upper gasket section **143**, the left gasket section **141**, and the right gasket section **142** in order to maintain contact against the skin supported by facial bones (e.g., nasal bone, zygomatic bone, mandible) of the user while maintaining glabrous cheek regions exposed within an interior volume of the shield center section **110**. Therefore, rather than the gasket **140** sliding across the face of the user during movement of the face (e.g., head turns, mandible movement during speaking) by the user, the gasket **140** can elastically deflect, thereby: maintaining contact between the upper gasket section **143** and the nasal bone of the user; and maintaining contact between the lower gasket section **144** and the bottom side of the mandible of the user.

5.2 Airflow Channel+Facial Cooling

[0052] In one implementation, the gasket **140** is configured to conform against the face of the user to: locate the filter assembly **120** offset a glabrous cheek region of the user; and form an airflow channel **150** extending from the filter assembly **120** to the transparent center region **112**, locating the nose and the mouth of the user, and configured to direct airflow across the glabrous cheek region of the user. In particular, the airflow channel **150**: defines a target depth range (e.g., between 0.1 inches and 0.3 inches, between 2.5 millimeters and 8 millimeters) between an interior surface of the shield center section **110** and the face of the user; and defines a target height range (e.g., between 0.5 inches and 1.5 inches, between 2.5 millimeters and 8 millimeters) across the glabrous cheek region of the user.

[0053] In one example, the gasket **140** is configured to conform against the face of the user to: locate the first filter assembly **120** offset a left glabrous cheek region of the user; and form a left airflow channel **150** extending from a first filter assembly **120** at the left filter window **114** to the transparent center region **112** of the shield center section **110** and configured to direct airflow—flowing within the left airflow channel **150**—across the left glabrous cheek region of the user. Similarly, the gasket **140** is configured to compress against the face of the user to: locate the second filter assembly **128** offset a right glabrous cheek region of the user; and form a right airflow channel **150** extending from the second filter assembly **128** at the right filter window **116** to the transparent center region **112** of the shield center section **110** and configured to direct airflow—flowing within the right airflow channel **150**—across the right glabrous cheek region of the user. Accordingly, air travels within the left airflow channel **150** across the left glabrous cheek region of the user and the right airflow channel **150** across the right glabrous cheek region of the user, thereby inducing a cooling effect for the user during inhalation and exhalation by the user.

[0054] For example, during inhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the first filter assembly **120** to the mouth of the user—laterally and vertically across the left glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the second filter assembly **128** to the mouth of the user—laterally and vertically across the right glabrous cheek region. Additionally, during exhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the mouth of the user to the first filter assembly **120**—laterally and vertically across the right glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the mouth of the user to the second filter assembly **128**—laterally and vertically across the right glabrous cheek region.

[0055] In another example, during inhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the first filter assembly **120** to the nose of the user—laterally and vertically across the left glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the second filter assembly **128** to the nose of the user—laterally and vertically across the right glabrous cheek region. Additionally, during exhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the nose of the user to the first filter assembly **120**—laterally and vertically across the right glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the nose of the user to the second filter assembly **128**—laterally and vertically across the right glabrous cheek region.

[0056] Therefore, the gasket **140** is configured to conform against the face of the user to form the airflow channel **150** within the interior volume of the shield center section **110** in order to induce a cooling effect across the glabrous cheek region of the user during inhalation (e.g., nasal inhalation, oral inhalation) and exhalation (e.g., nasal exhalation, oral exhalation) by the user.

5.3 Moisture Control

[0057] In one implementation, the shield **110** is formed of a transparent flexible material (e.g., polycarbonate film); and includes a hydrophilic coating arranged across an interior side and an exterior side of the lens center section **110**. Inhalation and exhalation by the user results in formation of moisture (e.g., fog, droplets) across the interior surface of the lens center section **110** which in turn obstructs facial cues (e.g., expressions) made by the user. Accordingly, the hydrophilic coating across the lens center section **110** prevents formation of moisture (e.g., fog, droplets) across the interior surface of the interior section; and directs moisture—by way of gravity—to lower gasket section **144** of the gasket **140** arranged about the shield center section **110**.

[0058] In another implementation, the gasket **140** can include a moisture trap **145** (e.g. cavity depression) molded into the lower section and cooperating with the hydrophilic coating and moisture shedding gasket skin surface to capture moisture formed within the interior side of the lens center section **110**. Thus, the gasket **140** enables the user to routinely (i.e., as needed) dispose of moisture collected within the moisture trap **145** molded into the lower gasket section **144** of the gasket **140** by the user manually breaking the gasket seal under and adjacent the trap area during any exhalation phase of the breathing cycle so as not to contaminate the plenum space with particulate. Other variations of this implementation can include a one-way release valve inlet at the base of the moisture trap **145** arranged at the lower gasket section **144** and configured to selectively release moisture accumulated within the moisture trap **145** at the lower gasket section **144** by the user performing a user seal check forcing the moisture trap valve to open providing a path for moisture collected in the trap to escape the respirator so as not to contaminate the plenum space with particulate.

6. Headband

[0059] Generally, the respirator apparatus **100** can include: a neck harness **160** coupled to the shield center section **110** and configured to wrap around a neck of the user; and a cranial harness **165** coupled to the shield center section **110** and configured to wrap about an upper cranium of the user. In particular, the cranial harness **165** cooperates with the neck harness **160** to: locate the shield center section **110** proximal the face of the user; and maintain conformance of the gasket **140** against the face of the user. Thus, during movement of the face by the user, the neck harness **160** cooperates with the cranial harness **165** to, rather than slide across the face of the user, maintain conformance of the gasket **140** against the face of the user.

[0060] In one implementation, the shield center section **110** includes: a lower left perforation **118** arranged proximal the left filter window **114** of the shield center section **110**; and a lower right perforation **118** arranged proximal the right filter window **116** of the shield center section **110** opposite the lower left perforation **118**. Additionally, the neck harness **160** includes: a first harness clip **162** (e.g., snuggler clip) arranged at a first end of the neck harness **160** and transiently coupled to the lower left perforation **118** of the lens center section **110**; and a second harness clip **162** (e.g., snuggler clip) arranged at a second end, opposite the first end, of the neck harness **160** and transiently coupled to the lower right perforation **118** of the lens center section **110**.

[0061] In one example, the neck harness **160** is formed of an elastomeric material (e.g., laminated foam backed micro-fiber, woven polyester flat webbing); and spans a target length (e.g., 26 inches, 66.04 centimeters) from the first end to the second end of the neck harness **160**; and spans a target height (e.g., 0.5 inches, 12 millimeters) configured to contour along a rear neck of the user.

[0062] Additionally, the shield center section **110** includes: an upper left perforation **118** arranged proximal the left filter window **114** of the shield center section **110**; and an upper right perforation **118** arranged proximal the right filter window **116** of the shield center section **110** opposite the upper left perforation **118**. Furthermore, the cranial harness **165** includes: a rear cranial section **166**—formed of the elastomeric material as described above configured to wrap about a rear cranium of the user; a third harness clip **162** (e.g., snuggler clip) arranged at a first end of the rear cranial section **166** and transiently coupled to the upper left perforation **118** of the shield center section **110**; and a fourth harness clip **162** (e.g., snuggler clip) arranged at a second end, opposite the first end, of the rear cranial section **166** and transiently coupled to the upper right perforation **118** of the shield center section **110**. The cranial harness **165** also includes a front cranial section **167** (e.g., thermoplastic strip); coupled the rear elastomeric cranial section **166**; and including an extended dogleg section oriented rearward on either side to wrap about a front cranium of the user; and cooperating with the rear cranial section to form a halo encircling an upper cranium of the user. This cranial harness provides the additional utility of setting the adjustment of the yoke for one hand respirator donning to the face of the user.

[0063] Accordingly, the cranial harness **165** cooperates with the neck harness **160** to locate the cranial harness **165** taut about the upper cranium of the user providing pressure to stabilize the gasket **140** against the face of the user. Therefore, during movement of the face by the user, the neck harness **160** cooperates with the neck harness **160** to maintain conformally the gasket **140** against the face of the user.

7. Respirator Setup+Facial Configuration

[0064] In one implementation, the respirator apparatus **100** includes the shield center section **110** including: the gasket **140** arranged about the rear side of the shield center section **110**; the transparent center region **112** configured to locate over the face of the user; the left filter window **114** laterally offset from the transparent center region **112**; and the right filter window **116** laterally offset from the transparent center region **112** opposite the left filter window **114**. During initial setup of the respirator apparatus **100**, the user can: couple a first filter assembly **120** within the left filter window **114** of the shield center section **110**; couple the second filter assembly **128** within the right filter window **116** of the shield center section **110**; couple the neck harness **160** to the shield center section **110**; and couple the cranial harness **165** to the shield center section **110**.

[0065] The neck harness **160** is configured to support the respirator apparatus **100** about the rear neck of the user and maintain the respirator apparatus **100** in a suspended configuration about the user's neck prior to placing the gasket **140** about the face of the user. The user can then: conform the gasket about the face of the user to induce direction of airflow through the filter assembly **120** into the interior volume of the shield center section **110**; and secure the

cranial harness **165** in a halo configuration about the upper cranium of the user. Accordingly, the cranial harness **165** cooperates with the neck harness **160** to maintain conformally the gasket **140** against the face of the user creating a sealed plenum between the shield and the face of the user. **[0066]** In the sealed configuration, the gasket **140** maintains conformally against the face of the user to form the left airflow channel **150** across the left glabrous cheek region of the user and the right airflow channel **150** across the right glabrous cheek region of the user. Accordingly, in the sealed configuration, the first filter assembly **120** is located offset the left glabrous cheek region of the user and the second filter assembly **128** is located offset the right glabrous cheek region of the user. Thus, during inhalation (e.g., nasal inhalation, oral inhalation) and exhalation (e.g., nasal exhalation, oral exhalation) the respirator apparatus **100** induces a cooling effect across the left glabrous cheek region and the right glabrous cheek region of the user.

[0067] Additionally, the gasket **140** cooperates with the face of the user such that—during movement of the face by the user—rather than slide across the face of the user, the gasket **140** elastically deforms while maintaining contact with the face of the user while directing airflow within the left airflow channel **150** and the right airflow channel **150**. **[0068]** Therefore, the respirator apparatus **100**: induces a cooling effect across the glabrous cheek region of the user during inhalation and exhalation of the user; enables an outside observer to view facial cues from the nose and the mouth region of the user located within the transparent center region **112** of the shield center section **110**; and prevents contaminants (e.g., airborne particulates, dirt, dust) about an exterior environment of the lens center section **110** from entering the interior of the lens (e.g., plenum) center section **110**.

[0069] As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

I claim:

1. A respirator apparatus comprising:

a shield:

configured to locate over a face of a user; and
defining a first filter window;

a first filter assembly comprising:

a first filter media; and

a first seal:

arranged about the first filter media; and

defining a first circumferential groove configured to mate with the first filter window to:

seal a first sealing surface of the first circumferential groove against an exterior surface of the first filter window;

seal a second sealing surface of the first circumferential groove against an interior surface, opposite the exterior surface, of the first filter window; and

transiently retain the first filter assembly within the first filter window; and

a gasket coupled to the shield and configured to:

conform against the face of the user; and

locate the first filter assembly offset from the face of the user to form a first airflow channel between the first filter assembly and the face of the user.

2. The respirator apparatus of claim 1:

wherein the shield defines:

a transparent center region; and

the first filter window offset from the transparent center region; and

wherein the gasket is configured to:

conform against the face of the user to align the center region with a nose and a mouth of the user; and

locate the first filter assembly offset from a first cheek region of the user to form the first airflow channel.

3. The respirator apparatus of claim 1:

wherein the first seal further comprises a first outer groove parallel to and offset from the first circumferential groove; and

wherein the first filter assembly is operable:

in a first configuration with the first circumferential groove mated to the first filter window to transiently retain the first filter assembly within the first filter window at a first depth from the face of the user; and

in a second configuration with the first outer groove mated to the first filter window to transiently retain the first filter assembly within the first filter window at a second depth, less than the first depth, from the face of the user.

4. The respirator apparatus of claim 3:

wherein the shield further defines a second filter window offset from the first filter window;

further comprising a second filter assembly:

comprising:

a second filter media; and

a second seal:

arranged about the second filter media;

defining a second circumferential groove; and

a second outer groove offset from the second circumferential groove;

operable in a third configuration to transiently retain the second filter assembly within the second filter window by mating the second circumferential groove with a second rim of the second filter window; and
operable in a fourth configuration to transiently retain the second filter assembly within the second filter window by mating the second outer groove with the second rim of the second filter window; and

wherein the gasket is further configured to locate the second filter assembly offset from a second cheek region of the user to form a second airflow channel.

5. The respirator apparatus of claim 1:

wherein the first seal further comprises a first outer groove parallel to and offset from the first circumferential groove; and

further comprising a filter cover comprising:

a pre-filter media comprising a first detent configured to couple to the first outer groove of the first seal to:

transiently retain the filter cover across an outer face of the first filter media; and

form a plenum between the outer face of the first filter media and an inner face of the pre-filter media.

6. The respirator apparatus of claim 1, wherein the first circumferential groove:

comprises:

the first sealing surface;

the second sealing surface arranged opposite the first sealing surface; and

a channel interposed between the first sealing surface and the second sealing surface; and
is configured to transiently couple to the first filter window to:

- locate a first rim of the first filter window of the shield within the channel;
- seal the first sealing surface against the exterior surface of the first rim of the first filter window; and
- seal the second sealing surface against the interior surface of the first rim of the first filter window, the second sealing surface cooperating with the first sealing surface to form an interference fit against the exterior surface and the interior surface of the first rim of the first filter window.

7. The respirator apparatus of claim 1, wherein the first filter assembly is configured to:

- during inhalation by the user:
 - direct air, flowing through the first filter media, laterally and vertically across the face of the user toward an orifice of the user; and
- during exhalation by the user:
 - direct air, exiting the orifice of the user, laterally and vertically across the face of the user toward the first filter media.

8. The respirator apparatus of claim 7:

wherein the shield further defines a second filter window laterally offset from the center region opposite the first filter window;

further comprising a second filter assembly comprising:
a second filter media; and
a second seal:

- arranged about the second filter media; and
- defining a second circumferential groove configured to mate with the second filter window to transiently retain the second filter assembly within the second filter window; and

wherein the gasket is further configured to locate the second filter assembly offset from a second cheek region of the user to form a second airflow channel between the second filter assembly and the second cheek region of the user.

9. The respirator apparatus of claim 1, wherein the first filter assembly is configured to:

- laterally and vertically deform about the first filter window during installation of the first filter assembly into the first filter window and mating of the first circumferential groove against the first filter window; and
- seal the first circumferential groove against a first rim of the first filter window following installation of the first filter assembly into the first filter window.

10. The respirator apparatus of claim 1:

wherein the shield further defines:

- a set of lower perforations; and
- a set of upper perforations; and

further comprising:

- a neck harness:
 - comprising a first set of clips configured to couple to the set of lower perforations; and
 - configured to wrap about a rear neck of the user; and
- a cranial harness:
 - comprising a second set of clips configured to couple to the set of upper perforations;
 - configured to wrap about a cranium of the user; and

configured to cooperate with the neck harness to locate the gasket conformally to the face of the user to form the first airflow channel.

11. The respirator apparatus of claim 1, wherein the gasket:

- comprises an elastomeric material; and
- is configured to conformally mate a sealing land to expose glabrous skin regions of the user and direct airflow from outside the shield through the first filter assembly across the glabrous skin regions into an internal volume of a shield plenum.

12. The respirator apparatus of claim 1, wherein the gasket comprises:

- a first gasket section:
 - arranged along a first lateral edge of the shield; and
 - configured to conform to a first side of a mandible of the user to locate the first filter window proximal a glabrous skin region of the user; and

a second gasket section:

- arranged along a second lateral edge, opposite the first lateral edge, of the shield;
- configured to conform to a second side, opposite the first side, of the mandible of the user; and
- cooperating with the first gasket section to, during movement of the mandible by the user, maintain contact between the gasket and the face of the user.

13. The respirator apparatus of claim 12, wherein the gasket further comprises:

an upper gasket section:

- arranged along an upper edge of a center region of the shield; and
- configured to conform against skin supported by a nasal bone of the user to locate the nose of the user within the center region of the shield; and

a lower gasket region:

- arranged along a lower edge of the center region of the shield;
- configured to conform to a lower side of the mandible of the user to locate a chin and the mouth of the user within the center region of the shield; and
- cooperating with the upper gasket section to elastically deflect, during movement of the mandible by the user, to:

- maintain contact between the upper gasket section and the skin on the nasal bone area of the user; and
- maintain contact between the lower gasket section and the skin on the lower side of the mandible of the user.

14. The respirator apparatus of claim 1:

wherein the shield comprises a hydrophilic coating arranged across an interior side of the shield; and

wherein the gasket comprises a moisture trap:

- molded into a lower section of the gasket; and
- cooperating with the hydrophilic coating to capture moisture formed within the interior side of the shield.

15. A respirator apparatus comprising:

a shield defining a filter window;

a filter assembly:

comprising:

- a filter media; and
- a seal:
 - arranged about the filter media;
 - defining a circumferential groove; and

defining an outer groove offset from the circumferential groove; and

operable in:

- a first configuration to transiently retain the filter assembly within the filter window by mating the circumferential groove to a rim of the filter window; and
- a second configuration to transiently retain the filter assembly within the filter window by mating the outer groove to the rim of the filter window; and

a gasket coupled to the shield and configured to conform against a face of the user to locate the filter assembly offset from the face of the user.

16. The respirator apparatus of claim **15**:
 wherein, in the first configuration, the circumferential groove mates with the rim of the filter window to transiently retain the filter assembly at a first depth from the face of the user; and
 wherein, in the second configuration, the outer groove mates with the rim of the filter window to transiently retain the filter assembly a second depth, less than the first depth, from the face of the user.

17. The respirator apparatus of claim **15**, wherein the circumferential groove:
 comprises:
 a first sealing surface;
 a second sealing surface arranged opposite the first sealing surface; and
 a channel interposed between the first sealing surface and the second sealing surface; and
 is configured to, in the first configuration, transiently couple to the filter window to:
 locate a rim of the filter window within the channel;
 seal the first sealing surface against an exterior surface of the rim of the filter window; and
 seal the second sealing surface against an interior surface, opposite the exterior surface, of the rim of the filter window, the second sealing surface cooperating with the first sealing surface to form an interference fit against the exterior surface and the interior surface of the rim of the filter window.

18. The respirator apparatus of claim **15**, further comprising a filter cover comprising:

- a pre-filter media comprising a detent configured to couple to the outer groove of the seal to:
 transiently retain the filter cover across an outer face of the filter media; and
- form a plenum between the outer face of the filter media and an inner face of the pre-filter media.

19. A respirator apparatus comprising:
 a shield defining a filter window;
 a filter assembly comprising:
 a filter media; and
 a seal:
 arranged about the filter media; and
 defining a circumferential groove configured to mate with the filter window to:
 seal opposing sealing surfaces of the circumferential groove against exterior surfaces and interior surfaces of a rim of the filter window;
 locate a rim of the filter window within a channel interposed between the opposing sealing surfaces; and
 transiently retain the filter assembly within the filter window; and
 a gasket coupled to the shield and configured to conform against the face of the user to locate the filter assembly offset from a face of the user.

20. The system of claim **19**:
 wherein the seal further comprises an outer groove offset from the circumferential groove; and
 wherein the filter assembly is operable:
 in a first configuration to transiently retain the filter assembly within the filter window by mating the circumferential groove with the rim of the filter window; and
 in a second configuration to transiently retain the filter assembly within the filter window by mating the outer groove with the rim of the filter window.

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